

Natalus Probe Card to NanoZ EK-IV DAQ Wiring and Assembly Manual

Technician-facing work instruction for soldering, connector wiring, continuity verification, and DAQ bring-up

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CONFIDENTIAL - Technician build package

Document control

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01	2026-07-30	Initial detailed Word manual prepared from supplied PowerPoint template, probe-card layout drawings, NanoZ pinout, and routing spreadsheets.	Engineering draft

Controlled-use note: This document is intended as a technician-facing build instruction. It should be checked against the physical probe card, breakout board silkscreen, ribbon-cable orientation, and NanoZ DAQ board labels before it is released as a production-controlled work instruction.

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1. Purpose and scope

This manual explains how the Natalus probe card, breakout board, ribbon-cable harness, and NanoZ EK-IV DAQ boards are connected for a 20-device parallel measurement row. It is written so that an engineering peer can hand it to a technician for soldering, wiring, continuity testing, and first-article bring-up.

The current probe card row measures 20 single devices in parallel. Each device has seven active contacts. Pads 1-4 are on the left side of each device and pads 5-7 are on the right side. The bottom-right pad is ground. The breakout board converts the probe-card ribbon-header signals into NanoZ 8-pin sensor-style connections.

This document does not define the wafer-level test recipe, NanoZ heater/sensor voltage settings, or production pass/fail limits. Those should be controlled in the ATA test recipe and NanoZ configuration file after electrical wiring has been verified.

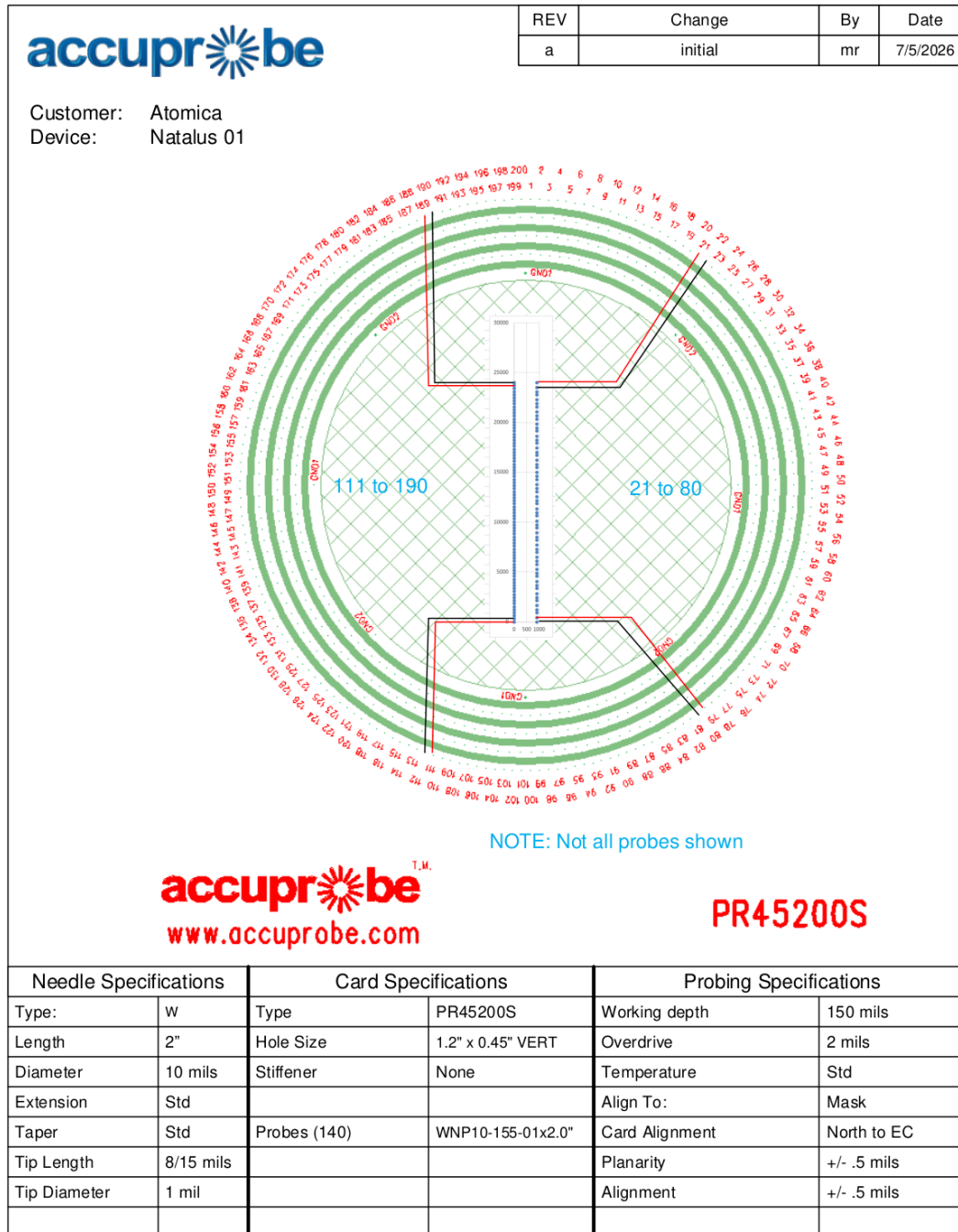
2. System overview - what is happening electrically

Electrical path: Probe needles contact a row of 20 devices on the wafer. The 140 active probe contacts route through the probe-card headers, through four ribbon cables, into the breakout board, and finally into NanoZ EK-IV DAQ board sensor inputs.

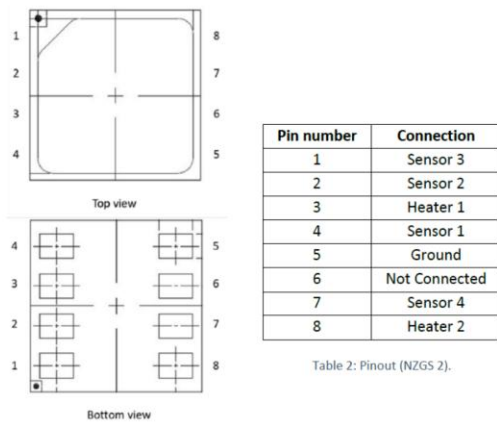
Stage	Physical item	Role in signal path	Technician focus
1	Wafer / device row	20 devices contacted simultaneously. Each device contributes 7 signals.	Do not probe until continuity and isolation checks pass.
2	Probe card	140 active probes arranged as two columns: left pads and right pads.	Confirm pin ranges and connector orientation.
3	4x 50-pin ribbon cables	Carries probe-card header signals to breakout board.	Label both ends and verify Pin 1 orientation.
4	Nautulus breakout board	Fans out the probe-card signals into per-device 8-pin NanoZ sensor headers.	Solder correct headers and inspect for bridges.
5	NanoZ EK-IV DAQ boards	Read sensor currents and drive heater lines according to NanoZ test settings.	Record EK S/N; connect by board identity, not COM port.

3. Source drawings and hardware references

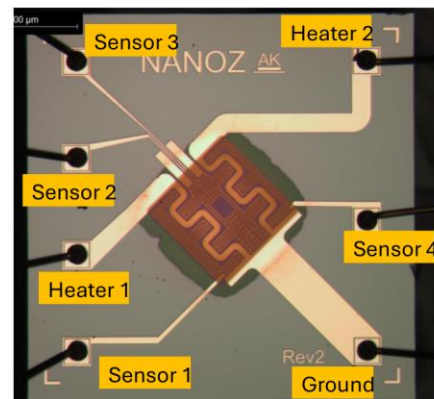
Keep the following drawings open at the bench during the first build. The probe-card drawing identifies the active pin ranges and probe-card specifications; the device pad layout identifies the 20-device row and seven-pad geometry; the NanoZ pinout identifies the final sensor/header function mapping.



Reference 1 - Natalus probe-card pin allocation and active ranges.

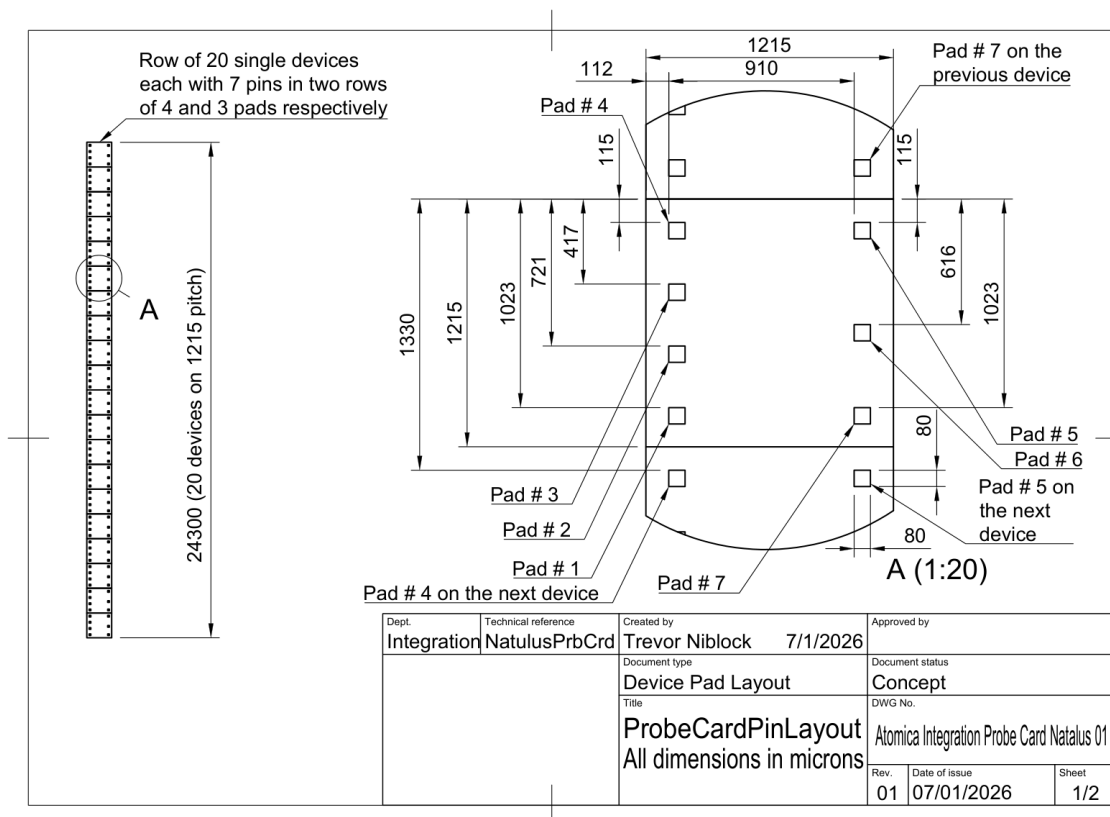


Pin connections NZGS 2. The chip comes in a 8-pin QFN package.



Optical image of NANOZ sensor

Reference 2 - NanoZ sensor pinout used as the endpoint wiring convention.



Reference 3 - Device pad layout: 20 devices on 1215 µm pitch with seven pads per device.

4. Connector and signal naming convention

Use this naming convention consistently on labels, traveler sheets, cable tags, and test data files.

Name used in this manual	Meaning
Device 01-20	The 20 devices contacted in one probe-card row.
P1-P7	Device pad number, from the probe-card/device layout.
B2-B8	Pin number on the breakout board 8-pin device header; B6 is intentionally not used by the NanoZ sensor pinout.
NanoZ pin 1-8	Pin number on the NanoZ sensor footprint/pinout. NanoZ pin 6 is not connected.
DAQ 01-10	NanoZ EK-IV DAQ boards actively used for 20 devices if two devices are connected per DAQ board.
DAQ 11-15	Reserved or future expansion boards for the requested 15-board system.

Source-data note: The probe-card source drawing lists active ranges 111 to 190 and 21 to 80. The routing spreadsheet maps those active contacts to the breakout board device headers and device pad numbers.

4.1 Universal device pad to NanoZ pin rule

Every one of the 20 device headers follows the same electrical rule. The technician should memorize this table and then use the per-device appendix table to find the correct source header pin for each device.

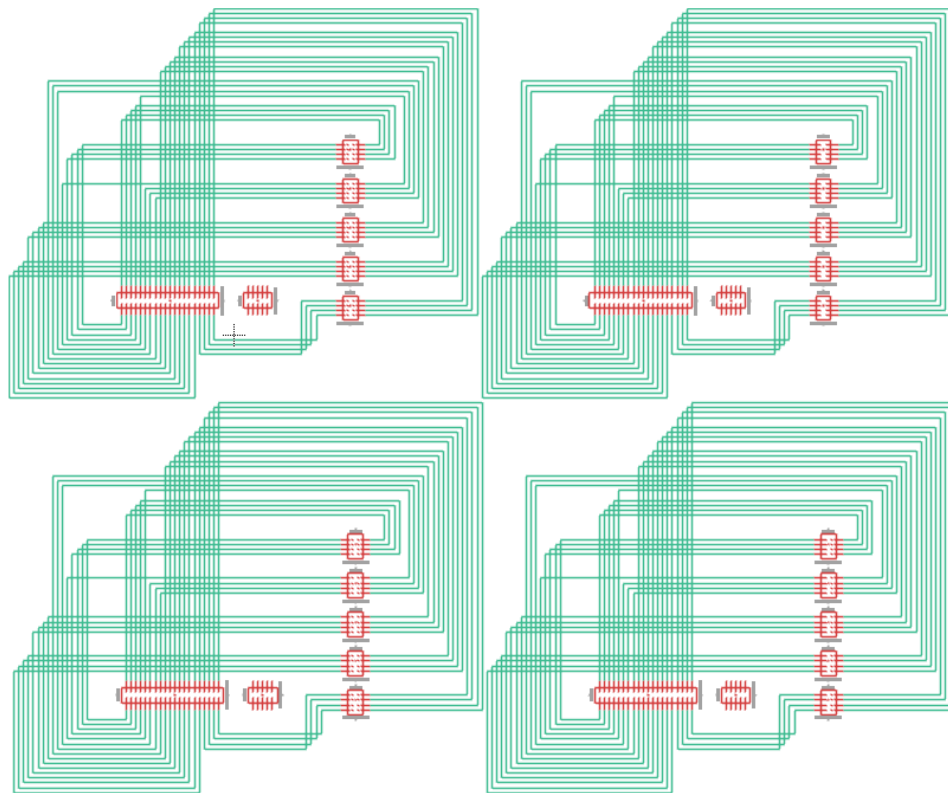
Device pad	Function	Probe-card side	NanoZ package pin	Breakout 8-pin header pin
1	Sensor 1	Left column	4	5
2	Heater 1	Left column	3	7
3	Sensor 2	Left column	2	8
4	Sensor 3	Left column	1	6
5	Heater 2	Right column	8	4
6	Sensor 4	Right column	7	2
7	Ground	Right column - bottom-right pad	5	3

Important: NanoZ pin 6 is not connected. Do not tie NanoZ pin 6 to ground or to any signal line. Ground is device pad 7, breakout pin B3, and NanoZ pin 5.

5. Breakout board population and soldering sequence

Populate the breakout board top-side connectors used by the probe-card fanout. The current BOM shows four 2x20 pin headers, four 2x5 pin headers, and twenty 2x4 / 8-position headers. The 8-position headers are the per-device NanoZ sensor outputs.

Reference designators	Qty	Description	Package	Value / MPN	Side
16	1	Board	Board	16	16
JP1, JP2, JP15, JP22	4	Pin Header	2X20	PINHD-2X20	Top
JP3, JP4, JP16, JP23	4	Pin Header	2X05	PINHD-2X5	Top
JP5, JP6, JP7, JP8, JP9, JP10, JP11, JP12, JP13, JP14, JP17, JP18, JP19, JP20, JP21, JP24, JP25, JP26, JP27, JP28	20	CONN HEADER VERT 8POS 1.27MM	62200821121	62200821121	Top



Breakout board routing view provided for connector-orientation reference.

5.1 Soldering order

1. Review the latest PCB silkscreen and verify that all connector reference designators match the released breakout-board drawing before soldering.
2. Clean the PCB and inspect all pads under magnification before placing connectors.
3. Install the lowest-profile headers first, then install taller pin headers. Keep all connectors square to the board.
4. Mark Pin 1 on every connector immediately after soldering using a paint pen or durable label.
5. Inspect all 1.27 mm and 0.1 inch header pins for solder bridges, insufficient wetting, lifted pads, or reversed orientation.
6. Do not connect any NanoZ DAQ board until the continuity and isolation traveler is completed.

6. Ribbon-cable and wiring procedure

The highest-risk assembly errors are left/right column swaps, flipped ribbon cables, and incorrect Pin 1 orientation. Use labels and meter checks rather than relying on cable color alone.

6.1 Cable labeling

7. Label each probe-card cable end with the probe-card connector name and Pin 1 direction.
8. Label the breakout-board end with the matching breakout input connector name and Pin 1 direction.
9. Add a second label showing which active range is carried by that cable: left-column range or right-column range.
10. Before tying down the harness, perform a spot-check from at least three known probe-card pins to their breakout endpoints.

6.2 Connection checks before final tie-down

Check	Method	Pass criterion
Pin 1 orientation	Visual check against drawing and silkscreen	Connector/cable Pin 1 agrees at both ends.

Left/right column identity	Continuity from known left pad and known right pad	Left pads route to P1-P4 signals; right pads route to P5-P7 signals.
Ground identification	Continuity from device P7 to breakout B3 / NanoZ pin 5	Ground line is continuous and not shorted to signal pins.
Unused NanoZ pin 6	Resistance from NanoZ pin 6 to all other pins	Open / floating.

7. NanoZ DAQ board assignment

Assign NanoZ DAQ boards by EK serial number and physical label, not by Windows COM port. COM ports may change after reconnecting USB cables; the NanoZ EK S/N should be treated as the stable hardware identity.

DAQ board	EK S/N / physical label	NanoZ input	Device header(s)	Status
DAQ 01		Chip 1 / Chip 2	Device 01 / Device 02	Active
DAQ 02		Chip 1 / Chip 2	Device 03 / Device 04	Active
DAQ 03		Chip 1 / Chip 2	Device 05 / Device 06	Active
DAQ 04		Chip 1 / Chip 2	Device 07 / Device 08	Active
DAQ 05		Chip 1 / Chip 2	Device 09 / Device 10	Active
DAQ 06		Chip 1 / Chip 2	Device 11 / Device 12	Active
DAQ 07		Chip 1 / Chip 2	Device 13 / Device 14	Active
DAQ 08		Chip 1 / Chip 2	Device 15 / Device 16	Active
DAQ 09		Chip 1 / Chip 2	Device 17 / Device 18	Active
DAQ 10		Chip 1 / Chip 2	Device 19 / Device 20	Active
DAQ 11		Reserved	Spare / future expansion	Reserved
DAQ 12		Reserved	Spare / future expansion	Reserved
DAQ 13		Reserved	Spare / future expansion	Reserved
DAQ 14		Reserved	Spare / future expansion	Reserved
DAQ 15		Reserved	Spare / future expansion	Reserved

Recommended first-article fill-in: Board 09 has already been confirmed as EK S/N 0002-0009 in Python testing. Use the same identify workflow for every board before final cable tie-down.

8. Continuity and isolation verification

Complete this section before connecting any NanoZ DAQ board to the breakout harness and before mounting the probe card on the prober.

Step	Test	Pass criterion	Initial / date
1	Visual inspection of all solder joints	No bridges, cold joints, or unsoldered pins	
2	Probe-card header to breakout input header continuity	Each mapped line < 2 Ω end-to-end	
3	Breakout sensor header to NanoZ cable continuity	Pins B2-B8 map to expected sensor functions	
4	Ground continuity	Every device ground pad ties to NanoZ ground only	
5	Isolation between neighboring non-ground signal lines	> 1 M Ω between adjacent non-ground pins	
6	Unused NanoZ pin 6 check	Pin 6 floating / NC	
7	DAQ board identify check	All connected boards return EK-IV signature 5164 and expected S/N	

9. DAQ bring-up and first-article test

Use a staged bring-up. The goal is to prove the wiring, NanoZ identity, data logging, and channel assignment before wafer probing.

11. Connect one NanoZ DAQ board by USB only and run the Python identify script. Record EK S/N, USB ID, and physical label.
12. Repeat for all DAQ boards before connecting them to the breakout board.
13. Connect the first DAQ board to Device 01 / Device 02 only. Perform continuity checks again with the DAQ cable installed.
14. Run a low-risk NanoZ identify/read test. Do not run heater drive or sensor bias until the correct NanoZ configuration has been reviewed.
15. Add one DAQ board at a time. After each addition, repeat identify, continuity, and isolation checks.
16. With the known-good sensor channel, run a short test cycle and confirm SPL and ENV CSV outputs are written.
17. Only after all channels pass, mount the probe card and run a dummy/engineering wafer row test before probing product wafers.

10. Technician traveler / sign-off sheets

Print this page with the appendix wiring tables and keep it with the assembly during soldering, continuity checks, and first-article bring-up.

Item	Technician sign-off	Date	Notes
Breakout PCB revision verified against released drawing			
All connectors installed and visually inspected			
Pin 1 marked on every connector and cable end			
Device 01-20 continuity checks complete			
Ground-only continuity verified for every device			
NanoZ pin 6 verified not connected			
DAQ S/N table completed			
First-article short run completed and data files reviewed			

Appendix A. Per-device wiring tables

Format: probe-card header pin -> breakout 8-pin device-header pin. The B-number is the breakout 8-pin header pin used by the NanoZ sensor cable. Verify the physical device-header reference designator on the latest breakout silkscreen before production release.

Dev	S1 P1	H1 P2	S2 P3	S3 P4	H2 P5	S4 P6	GND P7	BoB device header
01	J1-11 -> B5	J1-12 -> B7	J1-13 -> B8	J1-14 -> B6	J1-28 -> B4	J1-29 -> B2	J1-30 -> B3	Device header 01
02	J1-15 -> B5	J1-16 -> B7	J1-17 -> B8	J1-18 -> B6	J2-49 -> B4	J1-26 -> B2	J1-27 -> B3	Device header 02
03	J1-19 -> B5	J1-20 -> B7	J1-21 -> B8	J1-22 -> B6	J2-48 -> B4	J2-47 -> B2	J2-50 -> B3	Device header 03
04	J1-23 -> B5	J1-24 -> B7	J1-25 -> B8	J2-2 -> B6	J2-43 -> B4	J2-46 -> B2	J2-45 -> B3	Device header 04
05	J2-1 -> B5	J2-4 -> B7	J2-3 -> B8	J2-6 -> B6	J2-42 -> B4	J2-41 -> B2	J2-44 -> B3	Device header 05
06	J2-5 -> B5	J2-8 -> B7	J2-7 -> B8	J2-10 -> B6	J2-37 -> B4	J2-40 -> B2	J2-39 -> B3	Device header 06
07	J2-9 -> B5	J2-12 -> B7	J2-11 -> B8	J2-14 -> B6	J2-36 -> B4	J2-35 -> B2	J2-38 -> B3	Device header 07
08	J2-13 -> B5	J2-16 -> B7	J2-15 -> B8	J2-18 -> B6	J2-31 -> B4	J2-34 -> B2	J2-33 -> B3	Device header 08
09	J2-17 -> B5	J2-20 -> B7	J2-19 -> B8	J2-22 -> B6	J2-30 -> B4	J2-29 -> B2	J2-32 -> B3	Device header 09
10	J2-21 -> B5	J2-24 -> B7	J2-23 -> B8	J2-26 -> B6	J2-25 -> B4	J2-28 -> B2	J2-27 -> B3	Device header 10

Dev	S1 P1	H1 P2	S2 P3	S3 P4	H2 P5	S4 P6	GND P7	BoB device header
11	J3-1 -> B5	J3-2 -> B7	J3-3 -> B8	J3-4 -> B6	J3-48 -> B4	J3-49 -> B2	J3-50 -> B3	Device header 11
12	J3-5 -> B5	J3-6 -> B7	J3-7 -> B8	J3-8 -> B6	J3-45 -> B4	J3-46 -> B2	J3-47 -> B3	Device header 12
13	J3-9 -> B5	J3-10 -> B7	J3-11 -> B8	J3-12 -> B6	J3-42 -> B4	J3-43 -> B2	J3-44 -> B3	Device header 13
14	J3-13 -> B5	J3-14 -> B7	J3-15 -> B8	J3-16 -> B6	J3-39 -> B4	J3-40 -> B2	J3-41 -> B3	Device header 14
15	J3-17 -> B5	J3-18 -> B7	J3-19 -> B8	J3-20 -> B6	J3-36 -> B4	J3-37 -> B2	J3-38 -> B3	Device header 15
16	J3-21 -> B5	J3-22 -> B7	J3-23 -> B8	J3-24 -> B6	J3-33 -> B4	J3-34 -> B2	J3-35 -> B3	Device header 16
17	J3-25 -> B5	J4-2 -> B7	J4-1 -> B8	J4-4 -> B6	J3-30 -> B4	J3-31 -> B2	J3-32 -> B3	Device header 17
18	J4-3 -> B5	J4-6 -> B7	J4-5 -> B8	J4-8 -> B6	J3-27 -> B4	J3-28 -> B2	J3-29 -> B3	Device header 18
19	J4-7 -> B5	J4-10 -> B7	J4-9 -> B8	J4-12 -> B6	J4-50 -> B4	J4-49 -> B2	J3-26 -> B3	Device header 19
20	J4-11 -> B5	J4-14 -> B7	J4-13 -> B8	J4-16 -> B6	J4-45 -> B4	J4-48 -> B2	J4-47 -> B3	Device header 20

Appendix B. Full probe-card pin allocation reference

This table is generated from the supplied Natalus probe-card pin allocation spreadsheet. It is included for back-end debug, continuity-test tracing, and engineering review.

Devices 01-04

Dev	P	Function	Probe header pin	Land	X μ m	Y μ m	BoB hdr	BoB pin
1	1	Sensor1	J1-11	111	0	0	1	5
1	2	Heater 1	J1-12	112	0	302	1	7
1	3	Sensor 2	J1-13	113	0	606	1	8
1	4	Sensor 3	J1-14	114	0	908	1	6
1	5	Heater 2	J1-28	78	910	908	1	4
1	6	Sensor 4	J1-29	79	910	407	1	2
1	7	Ground	J1-30	80	910	0	1	3
2	1	Sensor1	J1-15	115	0	1215	2	5
2	2	Heater 1	J1-16	116	0	1517	2	7
2	3	Sensor 2	J1-17	117	0	1821	2	8
2	4	Sensor 3	J1-18	118	0	2123	2	6
2	5	Heater 2	J2-49	75	910	2123	2	4
2	6	Sensor 4	J1-26	76	910	1622	2	2
2	7	Ground	J1-27	77	910	1215	2	3
3	1	Sensor1	J1-19	119	0	2430	3	5
3	2	Heater 1	J1-20	120	0	2732	3	7
3	3	Sensor 2	J1-21	121	0	3036	3	8
3	4	Sensor 3	J1-22	122	0	3338	3	6
3	5	Heater 2	J2-48	72	910	3338	3	4
3	6	Sensor 4	J2-47	73	910	2837	3	2
3	7	Ground	J2-50	74	910	2430	3	3
4	1	Sensor1	J1-23	123	0	3645	4	5
4	2	Heater 1	J1-24	124	0	3947	4	7
4	3	Sensor 2	J1-25	125	0	4251	4	8
4	4	Sensor 3	J2-2	126	0	4553	4	6
4	5	Heater 2	J2-43	69	910	4553	4	4
4	6	Sensor 4	J2-46	70	910	4052	4	2
4	7	Ground	J2-45	71	910	3645	4	3



Devices 05-08

Dev	P	Function	Probe header pin	Land	X μ m	Y μ m	BoB hdr	BoB pin
5	1	Sensor1	J2-1	127	0	4860	5	5
5	2	Heater 1	J2-4	128	0	5162	5	7
5	3	Sensor 2	J2-3	129	0	5466	5	8
5	4	Sensor 3	J2-6	130	0	5768	5	6
5	5	Heater 2	J2-42	66	910	5768	5	4
5	6	Sensor 4	J2-41	67	910	5267	5	2
5	7	Ground	J2-44	68	910	4860	5	3
6	1	Sensor1	J2-5	131	0	6075	6	5
6	2	Heater 1	J2-8	132	0	6377	6	7
6	3	Sensor 2	J2-7	133	0	6681	6	8
6	4	Sensor 3	J2-10	134	0	6983	6	6
6	5	Heater 2	J2-37	63	910	6983	6	4
6	6	Sensor 4	J2-40	64	910	6482	6	2
6	7	Ground	J2-39	65	910	6075	6	3
7	1	Sensor1	J2-9	135	0	7290	7	5
7	2	Heater 1	J2-12	136	0	7592	7	7
7	3	Sensor 2	J2-11	137	0	7896	7	8
7	4	Sensor 3	J2-14	138	0	8198	7	6
7	5	Heater 2	J2-36	60	910	8198	7	4
7	6	Sensor 4	J2-35	61	910	7697	7	2
7	7	Ground	J2-38	62	910	7290	7	3
8	1	Sensor1	J2-13	139	0	8505	8	5
8	2	Heater 1	J2-16	140	0	8807	8	7
8	3	Sensor 2	J2-15	141	0	9111	8	8
8	4	Sensor 3	J2-18	142	0	9413	8	6
8	5	Heater 2	J2-31	57	910	9413	8	4
8	6	Sensor 4	J2-34	58	910	8912	8	2
8	7	Ground	J2-33	59	910	8505	8	3



Devices 09-12

Dev	P	Function	Probe header pin	Land	X μ m	Y μ m	BoB hdr	BoB pin
9	1	Sensor1	J2-17	143	0	9720	9	5
9	2	Heater 1	J2-20	144	0	10022	9	7
9	3	Sensor 2	J2-19	145	0	10326	9	8
9	4	Sensor 3	J2-22	146	0	10628	9	6
9	5	Heater 2	J2-30	54	910	10628	9	4
9	6	Sensor 4	J2-29	55	910	10127	9	2
9	7	Ground	J2-32	56	910	9720	9	3
10	1	Sensor1	J2-21	147	0	10935	10	5
10	2	Heater 1	J2-24	148	0	11237	10	7
10	3	Sensor 2	J2-23	149	0	11541	10	8
10	4	Sensor 3	J2-26	150	0	11843	10	6
10	5	Heater 2	J2-25	51	910	11843	10	4
10	6	Sensor 4	J2-28	52	910	11342	10	2
10	7	Ground	J2-27	53	910	10935	10	3
11	1	Sensor1	J3-1	151	0	12150	11	5
11	2	Heater 1	J3-2	152	0	12452	11	7
11	3	Sensor 2	J3-3	153	0	12756	11	8
11	4	Sensor 3	J3-4	154	0	13058	11	6
11	5	Heater 2	J3-48	48	910	13058	11	4
11	6	Sensor 4	J3-49	49	910	12557	11	2
11	7	Ground	J3-50	50	910	12150	11	3
12	1	Sensor1	J3-5	155	0	13365	12	5
12	2	Heater 1	J3-6	156	0	13667	12	7
12	3	Sensor 2	J3-7	157	0	13971	12	8
12	4	Sensor 3	J3-8	158	0	14273	12	6
12	5	Heater 2	J3-45	45	910	14273	12	4
12	6	Sensor 4	J3-46	46	910	13772	12	2
12	7	Ground	J3-47	47	910	13365	12	3



Devices 13-16

Dev	P	Function	Probe header pin	Land	X μ m	Y μ m	BoB hdr	BoB pin
13	1	Sensor1	J3-9	159	0	14580	13	5
13	2	Heater 1	J3-10	160	0	14882	13	7
13	3	Sensor 2	J3-11	161	0	15186	13	8
13	4	Sensor 3	J3-12	162	0	15488	13	6
13	5	Heater 2	J3-42	42	910	15488	13	4
13	6	Sensor 4	J3-43	43	910	14987	13	2
13	7	Ground	J3-44	44	910	14580	13	3
14	1	Sensor1	J3-13	163	0	15795	14	5
14	2	Heater 1	J3-14	164	0	16097	14	7
14	3	Sensor 2	J3-15	165	0	16401	14	8
14	4	Sensor 3	J3-16	166	0	16703	14	6
14	5	Heater 2	J3-39	39	910	16703	14	4
14	6	Sensor 4	J3-40	40	910	16202	14	2
14	7	Ground	J3-41	41	910	15795	14	3
15	1	Sensor1	J3-17	167	0	17010	15	5
15	2	Heater 1	J3-18	168	0	17312	15	7
15	3	Sensor 2	J3-19	169	0	17616	15	8
15	4	Sensor 3	J3-20	170	0	17918	15	6
15	5	Heater 2	J3-36	36	910	17918	15	4
15	6	Sensor 4	J3-37	37	910	17417	15	2
15	7	Ground	J3-38	38	910	17010	15	3
16	1	Sensor1	J3-21	171	0	18225	16	5
16	2	Heater 1	J3-22	172	0	18527	16	7
16	3	Sensor 2	J3-23	173	0	18831	16	8
16	4	Sensor 3	J3-24	174	0	19133	16	6
16	5	Heater 2	J3-33	33	910	19133	16	4
16	6	Sensor 4	J3-34	34	910	18632	16	2
16	7	Ground	J3-35	35	910	18225	16	3



Devices 17-20

Dev	P	Function	Probe header pin	Land	X μ m	Y μ m	BoB hdr	BoB pin
17	1	Sensor1	J3-25	175	0	19440	17	5
17	2	Heater 1	J4-2	176	0	19742	17	7
17	3	Sensor 2	J4-1	177	0	20046	17	8
17	4	Sensor 3	J4-4	178	0	20348	17	6
17	5	Heater 2	J3-30	30	910	20348	17	4
17	6	Sensor 4	J3-31	31	910	19847	17	2
17	7	Ground	J3-32	32	910	19440	17	3
18	1	Sensor1	J4-3	179	0	20655	18	5
18	2	Heater 1	J4-6	180	0	20957	18	7
18	3	Sensor 2	J4-5	181	0	21261	18	8
18	4	Sensor 3	J4-8	182	0	21563	18	6
18	5	Heater 2	J3-27	27	910	21563	18	4
18	6	Sensor 4	J3-28	28	910	21062	18	2
18	7	Ground	J3-29	29	910	20655	18	3
19	1	Sensor1	J4-7	183	0	21870	19	5
19	2	Heater 1	J4-10	184	0	22172	19	7
19	3	Sensor 2	J4-9	185	0	22476	19	8
19	4	Sensor 3	J4-12	186	0	22778	19	6
19	5	Heater 2	J4-50	24	910	22778	19	4
19	6	Sensor 4	J4-49	25	910	22277	19	2
19	7	Ground	J3-26	26	910	21870	19	3
20	1	Sensor1	J4-11	187	0	23085	20	5
20	2	Heater 1	J4-14	188	0	23387	20	7
20	3	Sensor 2	J4-13	189	0	23691	20	8
20	4	Sensor 3	J4-16	190	0	23993	20	6
20	5	Heater 2	J4-45	21	910	23993	20	4
20	6	Sensor 4	J4-48	22	910	23492	20	2
20	7	Ground	J4-47	23	910	23085	20	3

Appendix C. Open items before production release

Open item	Owner	Disposition needed
Confirm final active NanoZ board count: 10 active for 20 devices with 5 reserved, or a different 15-board wiring strategy.	Engineering	Update DAQ assignment table before production.
Freeze the probe-card-to-breakout connector orientation drawing.	Engineering	Add controlled figure with Pin 1 arrows for each cable.
Confirm physical device-header reference designators for Device 01-20 on the breakout-board silkscreen.	Engineering / Layout	Update Appendix A BoB header labels if different from Device Header 01-20.
Define production continuity traveler limits and release sign-off authority.	Quality / Engineering	Approve before technician build release.
Run first-article electrical test on known-good board 09 sensor channel.	Engineering	Attach CSV output and log to build package.

Appendix D. Source files used

Source file	How it was used
Natalus ProbeCard Pin allocation.pdf	Probe-card active range, card/probe specifications, visual active pin map.
ProbeCardPinLayout Drawing.pdf	20-device row layout, seven-pad geometry, pitch, and array pin numbering.
Pinout NANOZ Gas sensor.pdf	NanoZ sensor pin number to function mapping.
Natalus Probecard Pin Allocation.xlsx	Detailed routing table from probe-card header pins to breakout device-header pins.
Nautulus BreakoutBoard02.xlsx	Breakout board connector BOM and component references.
Atomica PPT Template 2025- UPDATED.pptx	Visual style, Atomica logo, confidentiality footer, and brand treatment.